

Lab Report: Study of Parallel Circuit and Determination of Current & Voltage

Experiment No

03

Objectives

- To study the behavior of a parallel circuit.
- To verify Ohm's Law in a parallel network.
- To determine total current, branch current and voltage.

Concept / Theory

In a parallel circuit, voltage across each branch remains the same while current divides among the branches depending on resistance. Ohm's Law states:

$$I = V / R$$

Apparatus

- DC Power Supply (5V)
- Resistors: 470Ω, 560Ω, 330Ω
- Ammeter
- Voltmeter
- Breadboard and connecting wires

Procedure

1. Connect the circuit as shown in the diagram.
2. Apply 5V DC supply.
3. Measure total current using ammeter.
4. Measure branch currents and voltage.
5. Record all observations carefully.

Observation Table

Voltage	Resistance (Theory)	Resistance (Practical)	Current (Theory)	Current (Practical)
5V	470Ω	460Ω	0.00737 A	0.0111 A
5V	560Ω	554Ω	0.00273 A	0.0028 A
5V	330Ω	326Ω	0.00464 A	0.0048 A

Calculations

Parallel Resistance:

$$R_p = (560 \times 330) / (560 + 330) = 207.64 \Omega$$

Total Resistance:

$$R = 470 + 207.64 = 677.64 \Omega$$

Total Current:

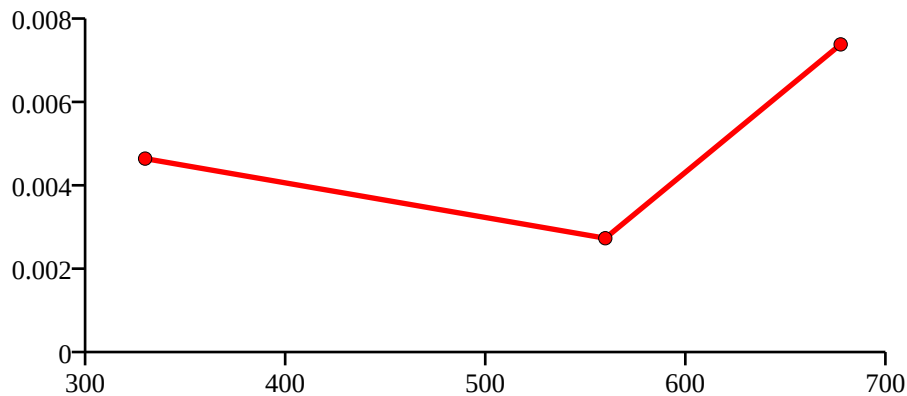
$$I = 5 / 677.64 = 0.00738 \text{ A}$$

Branch Currents:

$$i_1 = 1.53 / 560 = 0.00273 \text{ A}$$

$$i_2 = 1.53 / 330 = 0.00464 \text{ A}$$

Graph: Current vs Resistance



Error Analysis

Errors may occur due to resistor tolerance, loose connection, power supply fluctuation and instrument inaccuracies.

Result

The experiment successfully verified Ohm's Law and current division rule in a parallel circuit.

Precautions

- Ensure proper connections.
- Use correct resistor values.
- Avoid loose wiring.
- Check polarity before taking readings.

Discussion

Practical values differ slightly from theoretical values due to component tolerance and measuring instrument limitations.

References

- Basic Electrical Engineering Textbook
- Laboratory Manual
- Ohm's Law and Circuit Theory Notes